SAKSHAM TIMALSINA

University of Wyoming
Department of Civil and Architectural Engineering
and Construction Management

Phone: +1 307 761 2813
Email: sakshamtimalsina43@gmail.com
Personal Website

RESEARCH INTERESTS

Energy infrastructure safety and resilience, construction disputes and resolutions, construction safety, laser scanning, hydrogen energy, hydrogen transportation infrastructure, nuclear energy, and machine learning.

EDUCATION

University of Wyoming — Laramie, WY
Ph. D. Student, Civil Engineering — Expected: May 2028

Advisor: Dr. Chengyi Zhang

Tribhuvan University — Kathmandu, Nepal
B.S. in Civil Engineering — Sep 2022

Thesis: "*Flood Inundation Mapping and Dam Breach Analysis of Mugu-Karnali Basin using HEC-RAS*"

RESEARCH EXPERIENCE

University of Wyoming — Laramie, WY
Research Assistant (with Prof. Chengyi Zhang) — Jan 2024- present

- Application of both terrestrial and aerial laser scanners in the 3D model development, inspection, and maintenance of infrastructures.
- Hydrogen transportation infrastructure in Wyoming. Studied the relevance of different hydrogen infrastructures, including blending into natural gas pipelines, trucks, etc.
- Disputes in hydrogen pipelines and resolution through collaborative efforts and their prediction using machine learning models.

Tribhuvan University — Kathmandu, Nepal
Undergraduate Researcher — May 2021 – May 2022

- Water resources and disaster risk analysis through inundation mapping and dam breach analysis of Mugu-Karnali River segment using GIS and HEC-RAS.

TEACHING EXPERIENCE

Construction Documents, University of Wyoming — Jan-May 2026
- Grading, class assistance, and office hours

Introduction to Construction Management, University of Wyoming — Aug-Dec 2025
- Grading, office hours

Building Information Modelling, University of Wyoming — Jan-May 2025
- Grading, lab assistance, and office hours

Engineering Economics, University of Wyoming — Jan-May 2025
- Grading, office hours, and took 2 classes independently

Introduction to Civil, Architectural Engineering, and Construction Management, University of Wyoming Aug-Dec 2024

- Grading, office hours

ACADEMIC GRANTS/HONORS

- College of Engineering and Physical Sciences Travel Support Grant, 2025
- School of Graduate Education (SGE) Graduate Student Travel Award, 2025 and 2026
- Ranked in the top 250 (among 13,000 students) in the entrance exam by the Institute of Engineering, Nepal

PROFESSIONAL AFFILIATIONS AND REVIEW ACTIVITIES

- ASCE Student Member, American Society of Civil Engineers (ASCE)
- Reviewer for the Journal of Legal Affairs and Dispute Resolution in Engineering and Construction and Journal of Pipeline Systems Engineering and Practice, ASCE.
- Reviewer for the International Journal of Hydrogen Energy and Materials Today Sustainability, Elsevier.

INDUSTRY EXPERIENCE

Civil Engineering Intern Laramie, WY
DCR Consulting & Engineering May 2025- Aug 2025

- Preparation of updated drawings as per the field condition, grade design, and as-built drawings using AutoCAD, Civil3D, and Revit.
- Submittal registrar preparation and submittal review.
- Report generation and updates in P6.

Sub-engineer Sindhupalchowk, Nepal
Sunkoshi Rural Municipality Dec 2022- Nov 2023

- Design, estimate, and quality inspection of different civil engineering projects.

PUBLICATIONS

Journal papers

1. Zhao, Z., Kumar, D., Zhang, C.*, Li, H., **Timalsina, S.** (2024) "Techno-economic analysis of green hydrogen integration into existing pipeline infrastructure: A case study of Wyoming." *International Journal of Hydrogen Energy.* Volume 93, Pages 574-584, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2024.10.441
2. **Timalsina, S.**, Zhang, C.*, Kumar, D., Holubnyak, E. (2025). "Optimizing Hydrogen Transportation Infrastructure and Supply Chain under Uncertainty: An Integrated Mixed Integer Linear Programming and Real Options Analysis." *International Journal of Hydrogen Energy.* Volume 111, Pages 635-647, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2025.02.336
3. Pal, U. K., Zhang, C.*, and **Timalsina, S.** (2025). "Comparative Analysis of Regulatory Compliance and Legal Issues in Small Modular Nuclear Reactor Deployment." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction.* American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1270
4. **Timalsina, S.**, Zhang, C.*, Hada, S., Demirkesen, S. (2025). "Assessing Environmental, Safety, Regulatory, and Dispute Resolution Challenges in Hydrogen Pipeline Infrastructure." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction.* American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1272.
5. **Timalsina, S.**, Zhang, C.*, Quapp, U., Demirkesen, S., Olorunnishola, A. (2025). "Machine Learning-Based Predictive Model for Dispute Occurrence and Resolution Strategies in Pipeline

Projects." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction.* American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1334

6. Giri, M., Zhang, C.*, **Timalsina, S.**, Hada, S., Holubnyak, E. (2025). "A Comparative study of Regulatory and Policy Framework in Hydrogen Pipeline Development." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction*. American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1356
7. **Timalsina, S.**, Zhang, C.*, Pal, U. K. (2026) "Quantifying reliability and uncertainty in hydrogen infrastructure through integrated incident analysis." Journal of Loss Prevention in the Process Industries. Volume 100, 105921, ISSN 0950-4230, https://doi.org/10.1016/j.jlp.2026.105921

Conference papers

1. Mahajan, D., Sadikoglu, E., Pal, U. K., **Timalsina, S.**, Zhang, C.*, and Demirkesen, S. (2025). A Multi-Output Regression Approach to Predicting Gender-Centric Workplace Fatal Accidents. *Journal of Engineering, Project, and Production Management*, 15(1), 0002. DOI: https://doi.org/10.32738/JEPPM-2025-0002

CONFERENCE AND PRESENTATIONS

1. Attended *Construction Safety Conference*, Lawrence, KS, 2024 representing "A Multi-Output Regression Approach to Predicting Gender-Centric Workplace Fatal Accidents".
2. Presented at *Wyoming's Energy Future Symposium*, Laramie, WY, 2024: Poster title: "*Optimizing Hydrogen Transportation Infrastructure and Supply Chain under Uncertainty: An Integrated Mixed Integer Linear Programming and Real Options Analysis*".
3. Presented at *Graduate Poster Symposium*, University of Wyoming, WY, USA, 2024: Poster title: "*Optimizing Hydrogen Transportation Infrastructure and Supply Chain under Uncertainty: An Integrated Mixed Integer Linear Programming and Real Options Analysis*".
4. Presented at *CI Summit 2025,* Costa Mesa, CA, 2025. Presentation title: *"Assessing Environmental, Safety, Regulatory, and Dispute Resolution Challenges in Hydrogen Pipeline Infrastructure".*
5. Presented at *CI and CRC Summit 2026, San Antonio*, TX, 2026. Presentation title: *"Legal Risk Mitigation in Hydrogen Pipeline Transport through Corrective Action Order Analysis".*

WORKSHOPS ATTENDED

1. *AI Unlocked: Empowering Higher Education through Research and Discovery* — Denver, CO
   National Artificial Intelligence Research Resource (NAIRR) Pilot Workshop — Apr 2-3, 2025
2. *Urban Building Energy Modeling Workshop* — Laramie, WY
   University of Wyoming — Apr 29, 2025

CERTIFICATIONS AND LICENSURES

1. OSHA 30-Hour Outreach Training
2. Procore Certification: Student
3. Federal Aviation Administration (FAA) Part 107 License for small Unmanned Aerial Vehicle (sUAV)

LASER SCANNING PROJECTS INVOLVED

1. Laramie Railroad Depot — Apr, 2024
2. Cheyenne High Plains Arboretum — Jul, 2024
3. Fort Laramie — Jun, 2025

4. Cheyenne Capitol Old — Aug, 2025

GRANTS AND PROPOSALS SUBMITTED

1. Wyoming NASA EPSCoR Faculty Research Grant on "Machine Learning-Driven 3D Spatial Analytics for Post-Wildfire Infrastructure and Landscape Risk Assessment using Laser Scanning." — Submitted: Mar, 2026
2. American Heritage Center (AHC) 2026 Teaching & Research Grant on "A Research Symposium on the Digital Preservation of the Apatosaurus at the University of Wyoming." — Submitted: Feb, 2026
3. Coalitions & Collaboratives (COCO) Action, Implementation & Mitigation Grant on "Predictive Vegetation Growth Modeling Near Power Lines Using Laser Scanning for Wildfire Fuel Reduction." — Submitted: Oct, 2025
4. National Forest Foundation 2025 Collaborative Capacity Program - Mini Grants on "Building Regional Capacity for Digital Preservation of At-Risk Heritage Infrastructure in Medicine Bow National Forest." — Submitted: Jul, 2025
5. National Science Foundation Advanced Sensing and Computation for Environmental Decision-making (ASCEND) Engine in Colorado and Wyoming Request for Information (Nascent Phase, Wildfire in Power/Water Systems) 2025 — Submitted: Jul, 2025
6. National Science Foundation CO-WY Engine on "Assessing and Monitoring the Carbon Sequestration Potential of Forested Areas in Colorado and Wyoming Using Advanced LiDAR and Drone Technologies" FY 2024-25 — Submitted: Aug, 2024

SOFTWARE

Cyclone Register 360 Plus, Autodesk Revit, Primavera P6, Microsoft Project, AutoCAD, ArcGIS, Civil3D